

Decent Coaching Classes
VIII SB
Mind Your Math - VI

Subject : Mathematics (Ch. 4 to 8)

Marks : 30

Note : (i) All questions are compulsory subject-wise as per instructions. (ii) Draw neat and labelled figures wherever necessary.

Q.1) Solve any FIVE of the following :

- (1) The point of concurrence of the medians of a triangle is called the _____. [1]
- (2) Expand : $(x + 7)(x - 4)$ [1]
- (3) Factorise : $p^2 + 2p - 63$ [1]
- (4) If $a \propto 1/b$ and $a = 6$ when $b = 20$, find the constant of variation (k). [1]
- (5) If all sides of a quadrilateral are congruent, it is called a _____. [1]
- (6) State the property : Adjacent angles of a parallelogram are _____. [1]
- (7) If $x = ky$ and $x = 15$ when $y = 5$, find the value of k. [1]

Q.2) Solve any FOUR of the following :

- (1) Expand : $(2a - 5b)^2$ [2]
- (2) Factorise : $6x^2 + 5x - 6$ [2]
- (3) If $y \propto 1/x^2$ and $y = 8$ when $x = 3$, find y when $x = 6$. [2]
- (4) In rectangle ABCD, diagonals AC and BD intersect at P. If $l(AC) = 15$ cm, find $l(AP)$ and $l(PC)$. [2]
- (5) Find the length of the diagonal of a square whose side is 9 cm. [2]
- (6) G is the centroid of $\triangle ABC$. AD is the median through A. If $l(AG) = 10$ cm, find $l(GD)$ and $l(AD)$. [2]

Q.3) Solve any THREE of the following :

- (1) Expand : $(2x - 3y + 4)^2$ [3]
- (2) Factorise : $27a^3 + 8b^3$ [3]
- (3) Simplify : $(x^2 + 7x + 12) / (x^2 - 16) \div (x + 3) / (x - 4)$ [3]
- (4) Diagonals PR and QS of rhombus PQRS intersect at O. If $l(PR) = 24$ cm and $l(QS) = 18$ cm, find the side and perimeter of the rhombus. [3]
- (5) In parallelogram WXYZ, $m\angle W = 65^\circ$. Find the measures of $\angle X$, $\angle Y$ and $\angle Z$. [3]
- (6) 12 men can complete a piece of work in 18 days. In how many days will 18 men complete the same work? [3]

Q.4) Solve any TWO of the following :

- (1) Construct quadrilateral ABCD such that $l(AB) = 5$ cm, $l(BC) = 5.5$ cm, $l(CD) = 4$ cm, $l(AD) = 4.8$ cm and $l(AC) = 7$ cm. Write the steps of construction. [4]
- (2) Construct quadrilateral PQRS such that $l(PQ) = 4.5$ cm, $l(QR) = 5$ cm, $l(RS) = 4$ cm, $m\angle Q = 100^\circ$ and $m\angle R = 80^\circ$. Write the steps of construction. [4]
- (3) A car travelling at a uniform speed of 60 km/hr covers a certain distance in 5 hours. (i) Find the distance covered. (ii) If the same distance is to be covered in 4 hours, find the required speed. [4]
- (4) Simplify using expansion formulae : $(3x + 2y + z)^2 - (3x - 2y + z)^2$ [4]